

Unit-4

Multiple Integrals

Double
Integral

Triple
Integral

Double Integral

Let $f(x,y)$ be a function in 3D space with domain as xy -plane. Also let R be any region in xy -plane. Suppose we divide R in small sub-regions. Let $\Delta A_i = \Delta x_i \Delta y_i$ be the area of i th subregion. The double integral of $f(x,y)$ over the region R is defined as

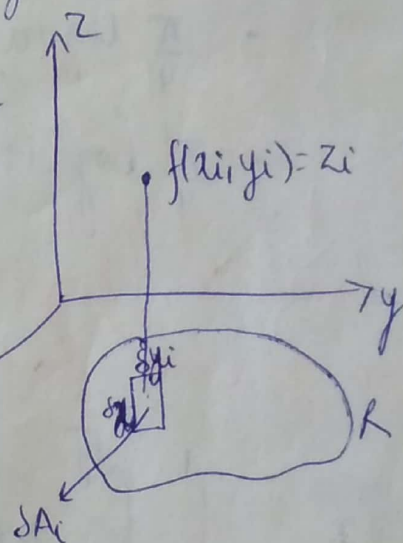
$$\iint_R f(x,y) dx dy = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i, y_i) \Delta x_i \Delta y_i$$

where (x_i, y_i) is any point in i th subregion.

Remark: (1) Double Integration $\iint_R f(x,y) dx dy$

gives the volume under the surface $f(x,y)$ and over the region R .

(2) If $f(x,y)=1$, then $\iint_R dx dy = \text{Area of region } R$.



Evaluation of Double Integral

Let R be bounded by the curve $y = f_1(x)$, $y = f_2(x)$ and ordinates $x=a$, $x=b$, then

$$\iint_R f(x,y) dx dy = \int_{x=a}^b \int_{y=f_1(x)}^{f_2(x)} f(x,y) dy dx$$

Note: If we take limits on x -axis, then we find limits of y by strip parallel to y -axis, and if we take limits on y -axis, then find limits of x -axis by strip parallel to x -axis.

Q. Evaluate $\int_0^1 \int_0^{\sqrt{1+x^2}} \frac{1}{1+x^2+y^2} dy dx$.

Solⁿ:

$$\begin{aligned}
 I &= \int_0^1 \int_0^{\sqrt{1+x^2}} \frac{1}{1+x^2+y^2} dy dx \\
 &= \int_0^1 \left[\int_0^{\sqrt{1+x^2}} \frac{1}{1+x^2+y^2} dy \right] dx \\
 &= \int_0^1 \frac{1}{\sqrt{1+x^2}} \tan^{-1} \left(\frac{y}{\sqrt{1+x^2}} \right) \Big|_{y=0}^{\sqrt{1+x^2}} dx \\
 &= \int_0^1 \frac{1}{\sqrt{1+x^2}} (\tan^{-1} 1 - \tan^{-1} 0) dx \\
 &= \int_0^1 \frac{\pi}{4\sqrt{1+x^2}} dx \\
 &= \frac{\pi}{4} \log(x + \sqrt{x^2+1}) \Big|_0^1 \\
 &= \frac{\pi}{4} \log(1 + \sqrt{2})
 \end{aligned}$$

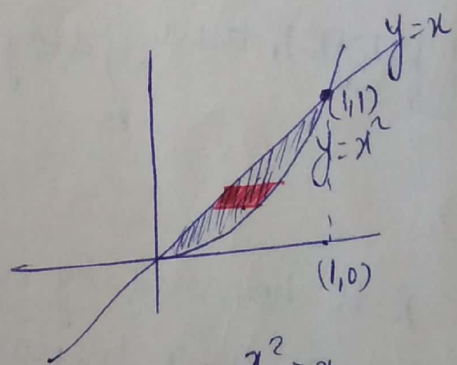
Q. Evaluate $\iint xy(x+y) dxdy$ over the region bounded by the line $y=x$ and the curve $y=x^2$.

$\therefore I = \int_{y=0}^1 \int_{x=y}^{\sqrt{y}} xy(x+y) dxdy$

$$= \int_{y=0}^1 \left(\frac{x^3 y}{3} + \frac{x^2 y^2}{2} \right) \Big|_{x=y}^{\sqrt{y}} dy$$

$$= \int_{y=0}^1 \left(\frac{y^{5/2}}{3} + \frac{y^3}{2} - \frac{5y^4}{6} \right) dy$$

$$= \left[\frac{y^{7/2}}{\frac{7}{2}} + \frac{y^4}{4} - \frac{5y^5}{30} \right]_0^1 = \frac{2}{21} + \frac{1}{4} - \frac{1}{6} = \frac{3}{56}$$



$$\begin{aligned}
 x^2 &= x \\
 \Rightarrow x^2 - x &= 0 \\
 \Rightarrow x(x-1) &= 0 \\
 \Rightarrow x &= 0, 1.
 \end{aligned}$$

Similarly it can be done as

$$I = \int_{x=0}^1 \int_{y=x^2}^x xy(x+y) dy dx$$

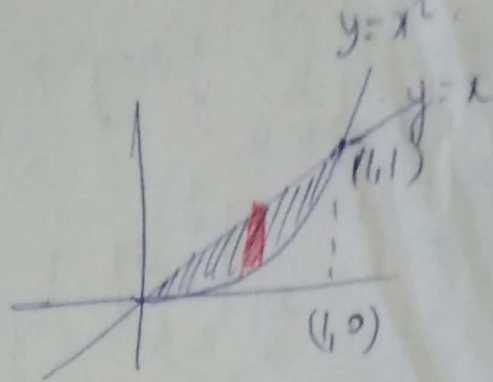
$$= \int_{x=0}^1 \left(\frac{xy^2}{2} + \frac{xy^3}{3} \right) \bigg|_{y=x^2}^x dx$$

$$= \int_{x=0}^1 \left(\frac{x^4}{2} + \frac{x^4}{3} - \frac{x^6}{2} - \frac{x^7}{3} \right) dx$$

$$= \frac{x^5}{10} + \frac{x^5}{15} - \frac{x^7}{14} - \frac{x^8}{24} \bigg|_0^1$$

$$= \frac{1}{10} + \frac{1}{15} - \frac{1}{14} - \frac{1}{24}$$

$$= \frac{3}{56}$$



Change of Order of Integration

Q. Change the order of integration and hence evaluate
 $\int_0^1 \int_x^{2-x} \frac{x}{y} dy dx$

$x = 2-x \rightarrow$ Point of intersection

$$2x = 2$$

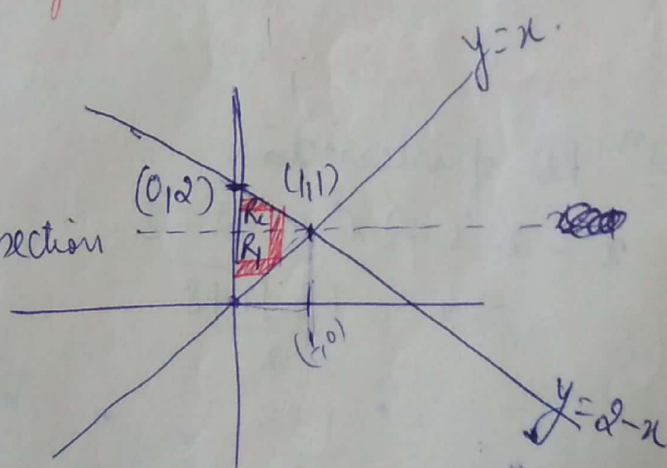
$$\Rightarrow x = 1$$

$$I = \int_0^1 \int_x^{2-x} \frac{x}{y} dy dx$$

$$= \iint_{R_1} \frac{x}{y} dx dy + \iint_{R_2} \frac{x}{y} dx dy$$

$$R_1: 0 \leq x \leq y, 0 \leq y \leq 1$$

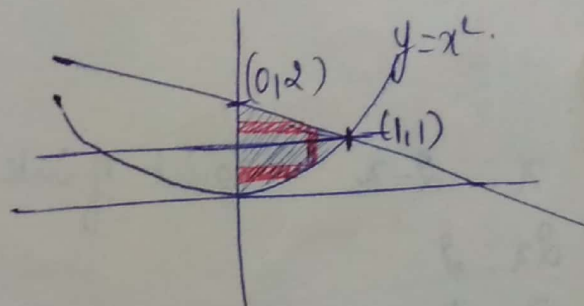
$$R_2: 0 \leq x \leq 2-y, 1 \leq y \leq 2$$



$$\begin{aligned}
 \therefore I &= \int_{y=0}^1 \int_{x=0}^y \frac{x}{y} dx dy + \int_{y=1}^2 \int_{x=0}^{2-y} \frac{x}{y} dx dy \\
 &= \int_{y=0}^1 \left. \frac{x^2}{2y} \right|_{x=0}^y dy + \int_{y=1}^2 \left. \frac{x^2}{2y} \right|_{x=0}^{2-y} dy \\
 &= \int_{y=0}^1 \frac{y^2}{2y} dy + \int_{y=1}^2 \frac{(2-y)^2}{2y} dy \\
 &= \left. \frac{y^2}{4} \right|_0^1 + \int_{y=1}^2 \frac{4+y^2-4y}{2y} dy \\
 &= \frac{1}{4} + \left[2 \log y + \frac{y^2}{4} - 2y \right]_{y=1}^2 \\
 &= \frac{1}{4} + \left[2 \log 2 + 1 - 4 - \frac{1}{4} + 2 \right] \\
 &= \frac{1}{4} + 2 \log 2 - \frac{1}{4} - 1 \\
 &= 2 \log 2 - 1.
 \end{aligned}$$

Q. Express $\int_0^1 \int_{x^2}^{2-x} f(x,y) dy dx$ as a sum of two double integrals.

Solⁿ: Pt. of intersection
 $x^2 = 2-x \Rightarrow x^2 + x - 2 = 0$
 $\Rightarrow x = \frac{-1 \pm \sqrt{1+8}}{2}$
 $= \frac{-1 \pm 3}{2}$
 $= 1, -2$

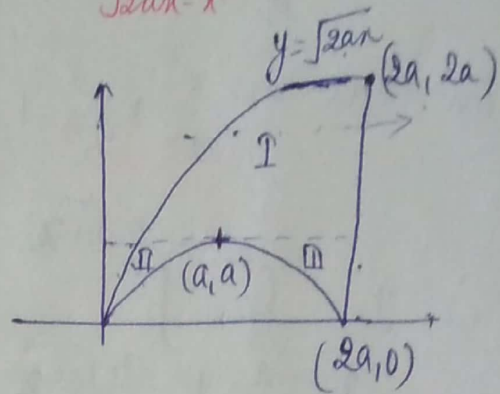


$$I = \int_{y=0}^1 \int_{x=0}^{\sqrt{y}} f(x,y) dx dy + \int_{y=1}^2 \int_{x=0}^{2-y} f(x,y) dx dy.$$

Change the order of integration in $\int_0^{2a} \int_{\sqrt{2ax-x^2}}^{2a} f(x,y) dy dx$.

Region 1:

$$I_1 = \int_{y=a}^{2a} \int_{x=y^2/2a}^{2a} f(x,y) dx dy$$



Region 2:

$$I_2 = \int_{y=0}^a \int_{x=y^2/2a}^{a-\sqrt{a^2-y^2}} f(x,y) dx dy$$

Region 3:

$$I_3 = \int_{y=0}^a \int_{x=a+\sqrt{a^2-y^2}}^{2a} f(x,y) dx dy$$

$$\therefore I = I_1 + I_2 + I_3$$

Double Integrals in Polar Coordinates

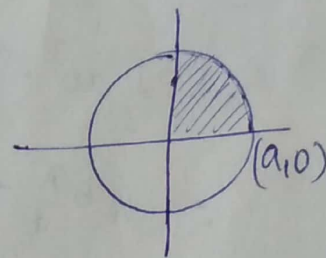
$$\iint_R f(x,y) dx dy \approx \iint_R f(r \cos \theta, r \sin \theta) r dr d\theta$$

Q. Change into polar coordinates and hence evaluate $\int_0^a \int_0^{\sqrt{a^2-x^2}} \sqrt{x^2+y^2} dy dx$.

Solⁿ:

$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases} \Rightarrow r^2 = x^2 + y^2$$

$$dx dy = r dr d\theta$$



$$\Rightarrow I = \int_{\theta=0}^{\pi/2} \int_{r=0}^a r \cdot r dr d\theta = \int_{\theta=0}^{\pi/2} \left. \frac{r^3}{3} \right|_0^a d\theta = \frac{a^3}{3} \times \frac{\pi}{2}$$

$$= \frac{\pi a^3}{6}$$

Q. Evaluate $\iint (x+y)^2 dx dy$ over the area bounded by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

Solⁿ: $I = \iint_A (x+y)^2 dx dy$

On A, i.e. on $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, $x = \pm a$
 $y = \pm \frac{b}{a} \sqrt{a^2 - x^2}$

$$\therefore I = \int_{x=-a}^a \int_{y=-\frac{b}{a}\sqrt{a^2-x^2}}^{\frac{b}{a}\sqrt{a^2-x^2}} (x^2 + y^2 + 2xy) dy dx$$

$$= 2 \int_{-a}^a \int_0^{\frac{b}{a}\sqrt{a^2-x^2}} (x^2 + y^2) dy dx$$

$$= 2 \int_{-a}^a \left[x^2 y + \frac{y^3}{3} \right]_{y=0}^{\frac{b}{a}\sqrt{a^2-x^2}} dx$$

$$= 2 \int_{-a}^a \left[\frac{1}{3} \frac{b^3}{a^3} (a^2 - x^2)^{3/2} + x^2 \frac{b}{a} \sqrt{a^2 - x^2} \right] dx$$

$$= 4 \int_0^a \left[\frac{b^3}{3a^3} (a^2 - x^2)^{3/2} + \frac{b}{a} x^2 (a^2 - x^2)^{1/2} \right] dx$$

$$= 4 \int_0^{\pi} \left[\frac{b^3}{3a^3} a^3 \cos^3 \theta + \frac{b}{a} a^2 \sin^2 \theta \cos \theta \right] a d\theta$$

$$= \frac{4}{3} ab^3 \frac{\left[\frac{1}{2}\right] \left[\frac{5}{2}\right]}{2\sqrt{3}} + 4a^3 b \frac{\left[\frac{3}{2}\right] \left[\frac{3}{2}\right]}{2\sqrt{3}}$$

$$= \frac{1}{3} ab^3 \cdot \frac{3}{2} \cdot \frac{1}{2} \left(\frac{1}{2}\right)^2 + a^3 b \cdot \frac{1}{2} \cdot \frac{1}{2} \left(\frac{1}{2}\right)^2$$

$$= a^3 b \frac{\pi}{4} + ab^3 \frac{\pi}{4}$$

$$= \frac{\pi}{4} ab(a^2 + b^2)$$

Multiple Integration

Cartesian coordinates

Q. Evaluate $\int_0^{\log 2} \int_0^x \int_0^{x+\log y} e^{x+y+z} \, dz \, dy \, dx$.

$$I = \int_0^{\log 2} \int_0^x \int_0^{x+\log y} e^{x+y} \cdot e^z \, dz \, dy \, dx$$

$$= \int_0^{\log 2} \int_0^x e^{x+y} \left. e^z \right|_0^{x+\log y} \, dy \, dx$$

$$= \int_0^{\log 2} \int_0^x e^{x+y} (e^{x+\log y} - 1) \, dy \, dx$$

$$= \int_0^{\log 2} \int_0^x (e^{2x+y+\log y} - e^{x+y}) \, dy \, dx$$

$$= \int_0^{\log 2} \left(y e^{2x} e^y - e^x e^y \right) \Big|_{y=0}^x \, dx$$

$$= \int_0^{\log 2} (x e^{3x} - e^{2x} - e^x) \, dx$$

$$= \frac{x e^{3x}}{3} - \int \frac{e^{3x}}{3} \, dx - \frac{e^{2x}}{2} - e^x \Big|_0^{\log 2}$$

$$= \frac{x e^{3x}}{3} - \frac{e^{3x}}{9} - \frac{e^{2x}}{2} - e^x \Big|_0^{\log 2}$$

$$= \frac{\log 2 \cdot e^{\log^3 2}}{3} - \frac{e^{\log^3 2}}{9} - \frac{e^{\log^2 2}}{2} - e^{\log 2} + \frac{1}{9} + \frac{1}{2} + 1$$

$$= \frac{8}{3} \log 2 - \frac{8}{9} - 2 - 2 + \frac{1}{9} + \frac{1}{2} + 1$$

$$= \frac{8}{3} \log 2 - \frac{7}{9} - 3 + \frac{1}{2}$$

$$= \frac{8}{3} \log 2 + \frac{-14-54+9}{18}$$

$$= \frac{8}{3} \log 2 - \frac{59}{18}$$

Q. Evaluate $\iiint_R \frac{dxdydz}{(x+y+z+1)^3}$ where R is the region bounded by the planes $x=0, y=0, z=0$ and $x+y+z=1$.

Soln:

$$\begin{aligned}
 I &= \int_0^1 \int_0^{1-x} \int_0^{1-x-y} \frac{1}{(x+y+z+1)^3} dz dy dx \\
 &= \int_0^1 \int_0^{1-x} \left[-\frac{1}{2} (x+y+z+1)^{-2} \right]_0^{1-x-y} dy dx \\
 &= -\frac{1}{2} \int_0^1 \int_0^{1-x} \left[(x+y+1-x-y+1)^{-2} - (x+y+1)^{-2} \right] dy dx \\
 &= -\frac{1}{2} \int_0^1 \int_0^{1-x} \left[\frac{1}{4} - \frac{1}{(x+y+1)^2} \right] dy dx \\
 &= -\frac{1}{2} \int_0^1 \left[\frac{1}{4} y + \frac{1}{x+y+1} \right]_0^{1-x} dx \\
 &= -\frac{1}{2} \int_0^1 \left(\frac{1-x}{4} + \frac{1}{x+1-x+1} - \frac{1}{1+x} \right) dx \\
 &= -\frac{1}{2} \int_0^1 \left(\frac{1-x}{4} + \frac{1}{2} - \frac{1}{1+x} \right) dx \\
 &= -\frac{1}{2} \left(\frac{x}{4} - \frac{x^2}{8} + \frac{x}{2} - \log(x+1) \right) \Big|_0^1 \\
 &= -\frac{1}{2} \left[\frac{1}{4} - \frac{1}{8} + \frac{1}{2} - \log 2 \right] \\
 &= -\frac{1}{2} \left[\frac{2-1+4}{8} - \log 2 \right] \\
 &= \frac{1}{2} \left[\log 2 - \frac{5}{8} \right]
 \end{aligned}$$

Cylindrical Polar Coordinates

The use of cylindrical coordinates r, θ, z is particularly suited for problems in which there is an axis of symmetry of solid. In place of volume element $dxdydz$ we use $rdrd\theta dz$.

If a denotes the radius of the base, h the altitude of a right circular cone, express its volume as a triple integral and evaluate it using cylindrical coordinates.

Solⁿ: From fig.

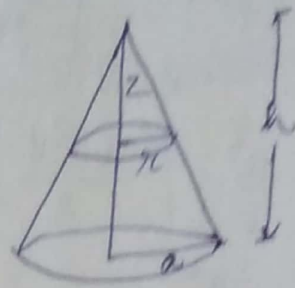
$$\frac{z}{r} = \frac{h}{a} \Rightarrow z = \frac{rh}{a}$$

$$V = \int_{\theta=0}^{2\pi} \int_{r=0}^a \int_{z=\frac{rh}{a}}^h r dr d\theta dz$$

$$= \int_0^{2\pi} \int_0^a \left[z \right]_{\frac{rh}{a}}^h r dr d\theta$$

$$= \int_0^{2\pi} \int_0^a \left(h r - \frac{h r^2}{a} \right) dr d\theta$$

$$= h \int_0^{2\pi} \left(\frac{a^2}{2} - \frac{a^3}{3a} \right) d\theta = \pi a^2 \frac{h}{3}$$



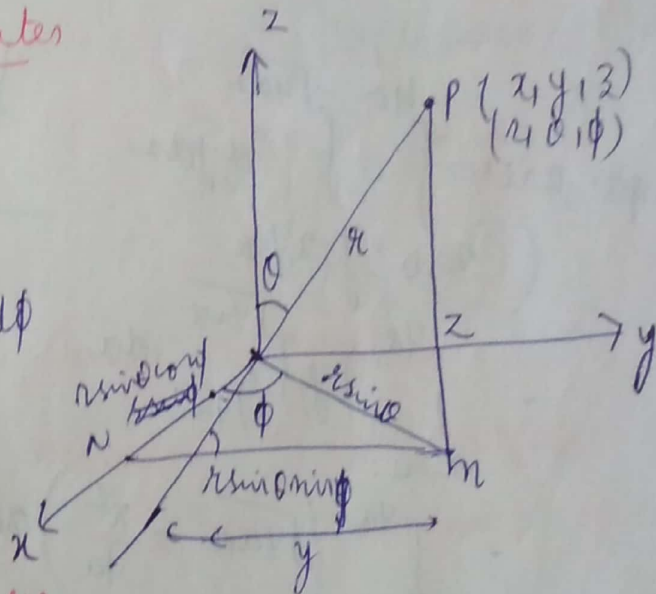
Spherical Polar Coordinates

$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

$$dv = r^2 \sin \theta dr d\theta d\phi$$



Q. Evaluate

$\iiint xyz dx dy dz$ over the volume in the positive octant of the sphere $x^2 + y^2 + z^2 = a^2$.

Solⁿ: $I = \iiint xyz dx dy dz$

$$= \int_{\theta=0}^{\pi/2} \int_{\phi=0}^{\pi/2} \int_{r=0}^a r \sin \theta \cos \phi \cdot r \sin \theta \sin \phi \cdot r \cos \theta \cdot r^2 \sin \theta dr d\theta d\phi$$

$$= \int_0^{\pi/2} \int_0^{\pi/2} \int_0^a r^5 \cos \phi \sin \phi \sin^3 \theta \cos \theta dr d\phi d\theta.$$

$$= \left[\frac{r^6}{6} \right]_0^a \int_0^{\pi/2} \cos \phi \sin \phi d\phi \int_0^{\pi/2} \sin^3 \theta \cos \theta d\theta.$$

$$= \frac{a^6}{6} \left[\frac{\sin^2 \phi}{2} \right]_0^{\pi/2} \left[\frac{\sin^4 \theta}{4} \right]_0^{\pi/2}$$

$$= \frac{a^6}{48}.$$

Applications of Double and Triple Integrals

I. Area by Double Integration

Q. Prove by method of double integration, that the area lying between the parabolas $y^2 = 4ax$ and $x^2 = 4ay$ is $\frac{16}{3}a^2$.

Solⁿ:

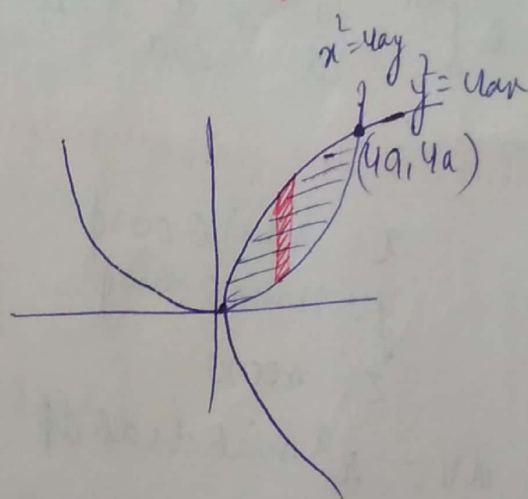
$$\text{Reqd. area} = \int_{x=0}^{4a} \int_{y=x^2/4a}^{\sqrt{4ax}} dy dx$$

$$= \int_{x=0}^{4a} \left[y \right]_{x^2/4a}^{\sqrt{4ax}} dx$$

$$= \int_0^{4a} \left(\sqrt{4ax} - \frac{x^2}{4a} \right) dx$$

$$= \left[\sqrt{4a} \times \frac{2}{3} x^{3/2} - \frac{x^3}{12a} \right]_0^{4a}$$

$$= \frac{16}{3} a^2.$$



Q. Find the area lying inside the circle $x^2 + y^2 - 2ax = 0$ and outside the circle $x^2 + y^2 = a^2$ using double integration.

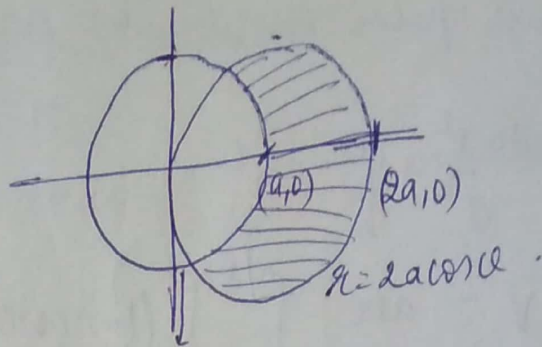
$$x^2 + y^2 = 2ax$$

$$\Rightarrow r^2 = 2a \cos \theta$$

$$\Rightarrow \boxed{r = 2a \cos \theta}$$

$$\text{and } x^2 + y^2 = a^2$$

$$\Rightarrow r^2 = a^2 \Rightarrow r = a \text{ in reqd region}$$



Point of intersection

$$2a \cos \theta = a \Rightarrow \cos \theta = \frac{1}{2} \Rightarrow \theta = \pm \frac{\pi}{3}$$

$$\begin{aligned} \therefore \text{Reqd area} &= 2 \times \int_0^{\pi/3} \int_a^{2a \cos \theta} r \, dr \, d\theta = 2 \times \int_0^{\pi/3} \left[\frac{r^2}{2} \right]_a^{2a \cos \theta} d\theta \\ &= \int_0^{\pi/3} (4a^2 \cos^2 \theta - a^2) d\theta \\ &= a^2 \int_0^{\pi/3} \{ 2(1 + \cos 2\theta) - 1 \} d\theta \\ &= a^2 \left[\theta + \frac{2 \sin 2\theta}{2} \right]_0^{\pi/3} \\ &= a^2 \left(\frac{\pi}{3} + \frac{\sqrt{3}}{2} \right) \end{aligned}$$

II. Volumes of solids by Double Integration

Q. Obtain the volume bounded by the surface $z = c \left(1 - \frac{x}{a}\right) \left(1 - \frac{y}{b}\right)$ and the quadrant of the elliptic cylinder $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

Soln: $V = \iint_C z \, dx \, dy$ ($\because \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ represents a cylinder whose axis is along z-axis and guiding curve is ellipse in solid geometry)

$$= \iint_C c \left(1 - \frac{x}{a}\right) \left(1 - \frac{y}{b}\right) dx \, dy$$

where $C: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

we use polar coordinates as $x = ar \cos \theta$, $y = br \sin \theta$
 $\Rightarrow dx dy = ab r dr d\theta$
 and $\frac{x^2}{a^2} + \frac{y^2}{b^2} = r^2$

$$\begin{aligned} \Rightarrow V &= abc \int_0^{\pi/2} \int_0^1 (1 - r \cos \theta)(1 - r \sin \theta) r dr d\theta \\ &= abc \int_0^{\pi/2} \int_0^1 \left[r - r^2 (\cos \theta + \sin \theta) + \frac{r^3}{2} \sin \theta \cos \theta \right] dr d\theta \\ &= abc \int_0^{\pi/2} \left[\frac{r^2}{2} - \frac{r^3}{3} (\cos \theta + \sin \theta) + \frac{r^4}{4} \sin \theta \cos \theta \right]_0^1 d\theta \\ &= abc \int_0^{\pi/2} \left[\frac{1}{2} - \frac{1}{3} (\cos \theta + \sin \theta) + \frac{1}{4} \sin \theta \cos \theta \right] d\theta \\ &= abc \left[\frac{\theta}{2} - \frac{1}{3} \sin \theta + \frac{1}{3} \cos \theta + \frac{1}{8} \sin^2 \theta \right]_0^{\pi/2} \\ &= abc \left(\frac{\pi}{4} - \frac{13}{24} \right) \end{aligned}$$

(III) volumes of solid as triple integral

Q. Find the volume of a sphere with centre at (a, b, c) and unit radius using spherical polar coordinates.

Solⁿ: Equation of sphere is $(x-a)^2 + (y-b)^2 + (z-c)^2 = 1$.
 Shifting the origin to (a, b, c) , equation of sphere becomes $x^2 + y^2 + z^2 = 1$.

$$\text{Volume} = \iiint_{\text{Sphere}} dx dy dz = \iiint dx dy dz.$$

Converting in spherical polar coordinates
 $x = r \sin \theta \cos \phi$ $y = r \sin \theta \sin \phi$ $z = r \cos \theta$

Here $r = 1$, $x = \sin \theta \cos \phi$ $y = \sin \theta \sin \phi$ $z = \cos \theta$

$$\text{qd. volume} = \int_0^1 \int_0^\pi \int_0^{2\pi} r^2 \sin\theta \, dr \, d\theta \, d\phi$$

$$= \int_0^\pi \int_0^{2\pi} \sin\theta \, d\theta \, d\phi \left(\frac{r^3}{3} \right)_0^1 = \frac{1}{3} \int_0^\pi \sin\theta \, d\theta \int_0^{2\pi} d\phi$$

$$= \frac{2\pi}{3} \int_0^\pi \sin\theta \, d\theta$$

$$= \frac{2\pi}{3} \left[-\cos\theta \right]_0^\pi$$

$$= \frac{4\pi}{3}$$

Vector Differentiation

Let $\vec{v}$ be a vector. $\vec{v}(t)$ is its value at $t = t_0$ if $\lim_{t \rightarrow t_0} \vec{v}(t) = \vec{v}(t_0)$.

Then, the derivative of $\vec{v}(t)$ is given by

$$\frac{d}{dt} \vec{v}(t) = \lim_{\delta t \rightarrow 0} \frac{\vec{v}(t + \delta t) - \vec{v}(t)}{\delta t}$$

General Rules of Vector Differentiation

For vectors $\vec{A}$, $\vec{B}$, $\vec{C}$, we have

1. $\frac{d}{dt} (\vec{A} \pm \vec{B}) = \frac{d\vec{A}}{dt} \pm \frac{d\vec{B}}{dt}$

2. $\frac{d}{dt} (u\vec{A}) = u \frac{d\vec{A}}{dt}$, u is a scalar function of t .

3. $\frac{d}{dt} (\vec{A} \cdot \vec{B}) = \frac{d\vec{A}}{dt} \cdot \vec{B} + \vec{A} \cdot \frac{d\vec{B}}{dt}$, and

$$\frac{d}{dt} (\vec{A} \times \vec{B}) = \frac{d\vec{A}}{dt} \times \vec{B} + \vec{A} \times \frac{d\vec{B}}{dt}$$

4. $\frac{d}{dt} (\vec{A} \cdot \vec{B} \times \vec{C}) = \frac{d\vec{A}}{dt} \cdot \vec{B} \times \vec{C} + \vec{A} \cdot \left(\frac{d\vec{B}}{dt} \times \vec{C} \right) + \vec{A} \cdot \left(\vec{B} \times \frac{d\vec{C}}{dt} \right)$

$$\frac{d}{dt} (\vec{A} \times (\vec{B} \times \vec{C})) = \frac{d\vec{A}}{dt} \times (\vec{B} \times \vec{C}) + \vec{A} \times \left(\frac{d\vec{B}}{dt} \times \vec{C} \right) + \vec{A} \times \left(\vec{B} \times \frac{d\vec{C}}{dt} \right)$$

* If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ denotes the position vector of a particle at any time t , then its velocity $\vec{v}$ and acceleration $\vec{a}$ are given by

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{dx}{dt} \hat{i} + \frac{dy}{dt} \hat{j} + \frac{dz}{dt} \hat{k}$$

and $\vec{a} = \frac{d^2x}{dt^2} \hat{i} + \frac{d^2y}{dt^2} \hat{j} + \frac{d^2z}{dt^2} \hat{k}$

Proofs

The necessary and sufficient condition for $\vec{v}(t)$ to have a constant magnitude is $\vec{v} \cdot \frac{d\vec{v}}{dt} = 0$.

Pf: first, let $\vec{v}$ has a constant magnitude.

$$\Rightarrow \vec{v} \cdot \vec{v} = |\vec{v}|^2 = \text{constant}$$

$$\Rightarrow \frac{d}{dt} (\vec{v} \cdot \vec{v}) = 0$$

$$\Rightarrow \frac{d\vec{v}}{dt} \cdot \vec{v} + \vec{v} \cdot \frac{d\vec{v}}{dt} = 0 \Rightarrow 2 \vec{v} \cdot \frac{d\vec{v}}{dt} = 0$$

$$\Rightarrow \vec{v} \cdot \frac{d\vec{v}}{dt} = 0$$

Conversely, if $\vec{v} \cdot \frac{d\vec{v}}{dt} = 0$

$$\Rightarrow \frac{d\vec{v}}{dt} \cdot \vec{v} + \vec{v} \cdot \frac{d\vec{v}}{dt} = 0$$

$$\Rightarrow \frac{d}{dt} (\vec{v} \cdot \vec{v}) = 0$$

$$\Rightarrow \vec{v} \cdot \vec{v} = \text{constant}$$

$$\Rightarrow |\vec{v}|^2 = \text{constant} \Rightarrow \vec{v} \text{ has constant modulus}$$

(2) the necessary and sufficient condition for any vector $\vec{v}$ to have a constant direction, is $\vec{v} \times \frac{d\vec{v}}{dt} = 0$.

Pf: first let $\vec{v}$

Q: A particle moves along a curve whose parametric equation in terms of time t , is $x = 3t^2$, $y = t^2 - 2t$, $z = t^3$. Find its velocity and acceleration at time $t = 2$.

Soln: $\vec{r} = \text{position vector} = x\hat{i} + y\hat{j} + z\hat{k}$
 $= 3t^2\hat{i} + (t^2 - 2t)\hat{j} + t^3\hat{k}$

velocity $\vec{v} = \frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j} + \frac{dz}{dt}\hat{k} = 6t\hat{i} + (2t - 2)\hat{j} + 3t^2\hat{k}$

$$\vec{v} \Big|_{t=2} = 12\hat{i} + 2\hat{j} + 12\hat{k}$$

$$\vec{a} = 6\hat{i} + 2\hat{j} + 6\hat{k}$$

$$\vec{a} \Big|_{t=2} = 6\hat{i} + 2\hat{j} + 12\hat{k}$$

Q. Find the angle between the tangents to the curve

$$\vec{r} = t^2\hat{i} - 2t\hat{j} + t^3\hat{k}$$

at the points $t=1$ and $t=2$.

Solⁿ: Tangent at any point 't' is given by

$$\frac{d\vec{r}}{dt} = 2t\hat{i} - 2\hat{j} + 3t^2\hat{k}$$

$\therefore$ the tangents at $t=1$ and $t=2$ are given by

$$\vec{T}_1 = 2\hat{i} - 2\hat{j} + 3\hat{k}$$

$$\text{and } \vec{T}_2 = 4\hat{i} - 2\hat{j} + 12\hat{k}$$

If θ is the angle b/w $\vec{T}_1$ and $\vec{T}_2$, then

$$\begin{aligned} \cos \theta &= \frac{\vec{T}_1 \cdot \vec{T}_2}{|\vec{T}_1| |\vec{T}_2|} = \frac{2(4) + (-2)(-2) + 3(12)}{\sqrt{2^2 + (-2)^2 + 3^2} \sqrt{4^2 + (-2)^2 + 12^2}} \\ &= \frac{48}{\sqrt{17} \sqrt{149}} \end{aligned}$$

$$\Rightarrow \theta = \cos^{-1} \left(\frac{24}{\sqrt{17} \sqrt{149}} \right)$$

Scalar/Vector Point Functions and Level Surfaces

A variable quantity which depends for its value on its position only, i.e. upon the coordinates, is called point function. There are two types of point functions:

(i) Scalar Point Function: If a scalar quantity ϕ depends on its position (x, y, z) in space, then $\phi(x, y, z)$ is called scalar point function. For a scalar point function ϕ ,

$$d\phi = \frac{\partial \phi}{\partial x} dx + \frac{\partial \phi}{\partial y} dy + \frac{\partial \phi}{\partial z} dz$$

Vector Point function: If a vector quantity $\vec{v}$ depends its value on its position (x, y, z) in space, then $\vec{v}(x, y, z)$ is called vector point function.

$$d\vec{v} = \frac{\partial \vec{v}}{\partial x} dx + \frac{\partial \vec{v}}{\partial y} dy + \frac{\partial \vec{v}}{\partial z} dz.$$

Level Surface

Let $\phi(x, y, z)$ be a scalar point function, continuous in a region. The surface drawn in space containing all those points where $\phi(x, y, z)$ has same value, is called level surface. e.g. equipotential surfaces.

Gradient of a Scalar Point function

The vector differential operator $\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \equiv \nabla$ operates on any scalar point function $\phi(x, y, z)$ and we get a vector quantity

$$\nabla \phi = \hat{i} \frac{\partial \phi}{\partial x} + \hat{j} \frac{\partial \phi}{\partial y} + \hat{k} \frac{\partial \phi}{\partial z}$$

is called gradient of the scalar point function ϕ .

- (*) The direction of $\nabla \phi$ is along the normal to the level surface.
The rate of change of ϕ is maximum in the direction of the normal.

Properties of Gradient:

- (i) $\nabla(u \pm v) = \nabla u \pm \nabla v$
- (ii) $\nabla(au) = a \nabla u$
- (iii) $\nabla(uv) = \nabla u \cdot v + u \cdot \nabla v$
- (iv) $\nabla[f(u)] = f'(u) \cdot \nabla u$
- (v) $\nabla(\vec{a} \cdot \vec{r}) = \vec{a}$; $\vec{a}$ is a constant vector.

Q. If $r^2 = x^2 + y^2 + z^2$, show that $\nabla f(r) = \frac{f'(r)}{r} \vec{r}$.
 Hence show that $\nabla\left(\frac{1}{r}\right) = -\frac{1}{r^3} \vec{r}$.

Solⁿ: $\nabla f(r) = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) f(r)$
 $= \hat{i} f'(r) \frac{\partial r}{\partial x} + \hat{j} f'(r) \frac{\partial r}{\partial y} + \hat{k} f'(r) \frac{\partial r}{\partial z}$
 $= f'(r) \left[\hat{i} \frac{\partial r}{\partial x} + \hat{j} \frac{\partial r}{\partial y} + \hat{k} \frac{\partial r}{\partial z} \right]$

~~$= f'(r) \vec{r}$~~

Now $r^2 = x^2 + y^2 + z^2$.

$\Rightarrow 2r \frac{\partial r}{\partial x} = 2x \Rightarrow \frac{\partial r}{\partial x} = \frac{x}{r}, \quad \frac{\partial r}{\partial y} = \frac{y}{r}, \quad \frac{\partial r}{\partial z} = \frac{z}{r}$

$\therefore \nabla f(r) = f'(r) \left[\hat{i} \frac{x}{r} + \hat{j} \frac{y}{r} + \hat{k} \frac{z}{r} \right]$
 $= \frac{f'(r)}{r} (x\hat{i} + y\hat{j} + z\hat{k})$
 $= \frac{f'(r)}{r} \vec{r}$

Taking $f(r) = \frac{1}{r} \Rightarrow \nabla\left(\frac{1}{r}\right) = -\frac{1}{r^2} \cdot \vec{r} = -\frac{1}{r^3} \vec{r}$

Q. Find a unit vector normal to the surface $x^2y + 2xz = 4$ at the point $(2, -2, 3)$.

Solⁿ: $\phi(x, y, z) = x^2y + 2xz - 4$

$\therefore \nabla\phi$ is normal to the surface

$\nabla\phi = \frac{\partial\phi}{\partial x} \hat{i} + \frac{\partial\phi}{\partial y} \hat{j} + \frac{\partial\phi}{\partial z} \hat{k}$
 $= (2xy + 2z) \hat{i} + x^2 \hat{j} + 2x \hat{k}$

At $(2, -2, 3)$, $\nabla\phi = (-8 + 6) \hat{i} + 4 \hat{j} + 4 \hat{k}$
 $= -2\hat{i} + 4\hat{j} + 4\hat{k}$

$\therefore$ Unit vector normal to surface $= \frac{-2\hat{i} + 4\hat{j} + 4\hat{k}}{\sqrt{4+16+16}} = \frac{-2\hat{i} + 4\hat{j} + 4\hat{k}}{6}$

2. Determine the constant 'a' such that at any point of intersection of the two spheres $(x-a)^2 + y^2 + z^2 = 3$ and $x^2 + (y-1)^2 + z^2 = 1$, their tangent planes are perpendicular to each other.

Let $\phi_1 = (x-a)^2 + y^2 + z^2 - 3$
 $\phi_2 = x^2 + (y-1)^2 + z^2 - 1$

Then $\nabla \phi_1 = 2(x-a)\hat{i} + 2y\hat{j} + 2z\hat{k}$

$\nabla \phi_2 = 2x\hat{i} + 2(y-1)\hat{j} + 2z\hat{k}$

The tangent planes will be perpendicular to each other when their normals are perpendicular to each other.

$\Rightarrow \nabla \phi_1 \cdot \nabla \phi_2 = 0$

$\Rightarrow (2(x-a)\hat{i} + 2y\hat{j} + 2z\hat{k}) \cdot (2x\hat{i} + 2(y-1)\hat{j} + 2z\hat{k}) = 0$

$\Rightarrow 4x(x-a) + 4y(y-1) + 4z^2 = 0$

$\Rightarrow x^2 - ax + y^2 - y + z^2 = 0$

$\Rightarrow x^2 + y^2 + z^2 - ax - y = 0 \quad \text{--- (1)}$

At the point of intersection of given spheres

$(x-a)^2 + y^2 + z^2 - 3 = x^2 + (y-1)^2 + z^2 - 1 = 0$

Adding both we get,

$x^2 + y^2 + z^2 - ax - y + \frac{1}{2}a^2 - \frac{3}{2} = 0 \quad \text{--- (2)}$

Comparing (1) and (2), we get

$\frac{a^2}{2} - \frac{3}{2} = 0 \Rightarrow a = \pm \sqrt{3}$

Directional Derivatives

The directional derivative of ϕ in the direction $\vec{a}$ is $\nabla\phi \cdot \hat{a}$.

Q. Compute the directional derivative of $x^2 + y^2 + 4xz$ at $(1, -2, 2)$ in the direction of the vector $2\hat{i} - 2\hat{j} - \hat{k}$.

Solⁿ: Let $f = x^2 + y^2 + 4xz$

$$\nabla f = (2x + 4z)\hat{i} + 2y\hat{j} + 4x\hat{k}$$

$$\nabla f|_{(1, -2, 2)} = 10\hat{i} - 4\hat{j} + 4\hat{k}$$

Directional derivative of f at $(1, -2, 2)$ along the vector $2\hat{i} - 2\hat{j} - \hat{k}$

$$\approx \nabla f \cdot \frac{(2\hat{i} - 2\hat{j} - \hat{k})}{\sqrt{4+4+1}}$$

$$= \frac{20 + 8 - 4}{3}$$

$$= \frac{24}{3} = 8$$

Q. What is the directional derivative of $\phi = xy^2 + yz^3$ at $(2, -1, 1)$ in the direction of normal to the surface $x \log z - y^2 = -4$ at $(-1, 2, 1)$?

Solⁿ: $\nabla\phi = y^2\hat{i} + (2xy + z^3)\hat{j} + 3yz^2\hat{k}$

$$\nabla\phi|_{(2, -1, 1)} = \hat{i} - 3\hat{j} - 3\hat{k}$$

Let other surface be $\phi_1 = x \log z - y^2 + 4$.

$$\nabla\phi_1 = \log z \hat{i} - 2y\hat{j} + \frac{x}{z}\hat{k}$$

$$\nabla\phi_1|_{(-1, 2, 1)} = 0\hat{i} - 4\hat{j} - \hat{k} = \vec{N} \text{ (say)}$$

direction

directional derivative of ϕ along $\hat{n}$ is

$$\nabla\phi \cdot \hat{n} = \hat{i} - 3\hat{j} - 3\hat{k} \cdot \frac{(-4\hat{j} - \hat{k})}{\sqrt{16+1}}$$

$$= \frac{15}{\sqrt{17}}$$

Divergence of a Vector Point Function

When the operator ∇ operates scalarly on a vector point function, we call it is a divergence.

If $\vec{F}(x, y, z)$ is a vector point function, then

$$\nabla \cdot \vec{F} = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot (F_1 \hat{i} + F_2 \hat{j} + F_3 \hat{k})$$

$$= \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$$

$$= \text{div } \vec{F}$$

If $\nabla \cdot \vec{F} = 0$, $\vec{F}$ is called SOLENOIDAL

Q. Find the value of λ for which $\vec{u} = (x+2y)\hat{i} + (y-2z)\hat{j} + (x+\lambda z)\hat{k}$ is solenoidal.

Solⁿ: For $\vec{u}$ to be solenoidal, $\nabla \cdot \vec{u} = 0$

$$\Rightarrow \frac{\partial}{\partial x}(x+2y) + \frac{\partial}{\partial y}(y-2z) + \frac{\partial}{\partial z}(x+\lambda z) = 0$$

$$\Rightarrow 1 + 1 + \lambda = 0 \Rightarrow \lambda = -2$$

Q. If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, show that $\nabla(r^n \cdot \vec{r}) = (n+3)r^n$.

$$\text{Solⁿ: } \nabla(r^n \cdot \vec{r}) = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) (r^n x \hat{i} + r^n y \hat{j} + r^n z \hat{k})$$

$$= \frac{\partial}{\partial x}(r^n x) + \frac{\partial}{\partial y}(r^n y) + \frac{\partial}{\partial z}(r^n z)$$

$$= r^n \frac{\partial x}{\partial x} + n r^{n-1} \frac{\partial r}{\partial x} \cdot x + r^n \cdot 1 + y n r^{n-1} \frac{\partial r}{\partial y} + r^n \cdot 1 + z n r^{n-1} \frac{\partial r}{\partial z}$$

$$= 3n^n + n^{n-1} \left(x \cdot \frac{x}{n} + y \cdot \frac{y}{n} + z \cdot \frac{z}{n} \right) \\ = (n+3)n^n.$$

Curl of a vector field

When a vector point function is operated vectorially by ∇ , result is a vector quantity called curl.

If $\vec{F} = f_1 \hat{i} + f_2 \hat{j} + f_3 \hat{k}$ is a vector point function, then

$$\text{curl } \vec{F} = \nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f_1 & f_2 & f_3 \end{vmatrix}$$

(*) $\vec{F}$ is called irrotational if $\nabla \times \vec{F} = 0$. If $\vec{F}$ is irrotational, a scalar point function ϕ exists such that $\vec{F} = \nabla \phi$, ϕ is called scalar potential.

Q. If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ and $\vec{a}$ is a constant vector, show that $\vec{r}$ is irrotational.

Solⁿ: $\text{curl } \vec{r} = \nabla \times \vec{r} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x & y & z \end{vmatrix}$

$$= \hat{i} \left(\frac{\partial z}{\partial y} - \frac{\partial y}{\partial z} \right) - \hat{j} \left(\frac{\partial z}{\partial x} - \frac{\partial x}{\partial z} \right) + \hat{k} \left(\frac{\partial y}{\partial x} - \frac{\partial x}{\partial y} \right)$$

$$= 0$$

$\therefore \vec{r}$ is irrotational.

Q. A vector field $\vec{F} = (x^2 - y^2 + z)\hat{i} - (2xy + y)\hat{j}$. Show that the field is irrotational and find its scalar potential.

Solⁿ: $\text{curl } \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 - y^2 + z & -2xy - y & 0 \end{vmatrix} = 0 \Rightarrow \vec{F} \text{ is irrotational}$

obtain its scalar potential ϕ , we have $\vec{F} = \nabla\phi$
 ~~$x^2 - y^2 + x$~~ $(x^2 - y^2 + x)\hat{i} - (2xy + y)\hat{j} = \frac{\partial\phi}{\partial x}\hat{i} + \frac{\partial\phi}{\partial y}\hat{j} + \frac{\partial\phi}{\partial z}\hat{k}$

$$\Rightarrow \frac{\partial\phi}{\partial x} = x^2 - y^2 + x, \quad \frac{\partial\phi}{\partial y} = -2xy - y \quad (*)$$

Integrating w.r.t x

$$\phi = \frac{x^3}{3} - y^2x + \frac{x^2}{2} + C(y)$$

$$\frac{\partial\phi}{\partial y} = -2yx + \frac{\partial C}{\partial y} \quad \text{comparing with } (*)$$

$$\frac{\partial C}{\partial y} = -y \Rightarrow C = -\frac{y^2}{2}$$

$$\therefore \phi = \frac{x^3}{3} - xy^2 + \frac{x^2}{2} - \frac{y^2}{2}$$

Q. Show that $\vec{F} = (2xy + z^3)\hat{i} + x^2\hat{j} + 3z^2x\hat{k}$ is conservative field. Find its scalar potential and also the work done in moving a particle from $(1, -2, 1)$ to $(3, 1, 4)$.

sgn: $\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xy + z^3 & x^2 & 3z^2x \end{vmatrix} = 0 \Rightarrow \vec{F} \text{ is conservative (irrotational)}$

$\Rightarrow$ a scalar potential ϕ st $\vec{F} = \nabla\phi$.

$$\Rightarrow \frac{\partial\phi}{\partial x} = 2xy + z^3, \quad \frac{\partial\phi}{\partial y} = x^2, \quad \frac{\partial\phi}{\partial z} = 3z^2x$$

Integrating w.r.t x

$$\phi = x^2y + xz^3 + f(y, z)$$

$$\Rightarrow \frac{\partial\phi}{\partial y} = x^2 + \frac{\partial f}{\partial y} \Rightarrow \frac{\partial f}{\partial y} = 0 \Rightarrow f = g(z)$$

$$\text{Next, } \frac{\partial\phi}{\partial z} = 3z^2x + \frac{\partial g}{\partial z} \Rightarrow \frac{\partial g}{\partial z} = 0 \Rightarrow g = C$$

$$\therefore \phi = x^2y + xz^3 + C$$

$$\text{work done} = \int_C \vec{F} \cdot d\vec{r} = \phi_B - \phi_A$$

$$\left[\phi(3, 1, 4) - \phi(1, -2, 1) \right] = \left[x^2y + xz^3 + C \right]_{(1, -2, 1)}^{(3, 1, 4)}$$

$$= 202$$

Vector Integration

Line Integral

$$\int_C \vec{F} \cdot d\vec{r}, \quad d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}$$

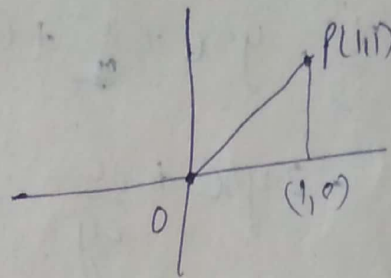
Q. Let the vector field be $\vec{F} = x^2\hat{i} - xy\hat{j}$. Find its line integral from the origin O to $P(1,1)$.

(i) along the straight line OP

(ii) along $y^2 = x$

Solⁿ: (i)

$$\begin{aligned} \int_{OP} \vec{F} \cdot d\vec{r} &= \int_{OP} (x^2\hat{i} - xy\hat{j}) (dx\hat{i} + dy\hat{j}) \\ &= \int_{OP} (x^2 dx - xy dy) \end{aligned}$$



on $OP: y=x \Rightarrow dy=dx \Rightarrow \int_{OP} \vec{F} \cdot d\vec{r} = \int_{x=0}^1 x^2 dx - x^2 dx = 0$

(ii) $y^2 = x \Rightarrow 2y dy = dx$
 $\Rightarrow y dy = \frac{dx}{2}$

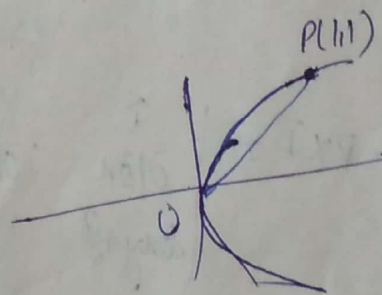
$$\therefore \int_{\text{curve } OP} \vec{F} \cdot d\vec{r} = \int_{OP} (x^2 dx - xy dy)$$

$$= \int_{x=0}^1 x^2 dx - \int_0^1 \frac{x}{2} dx$$

$$= \left. \frac{x^3}{3} \right|_0^1 - \left. \frac{x^2}{4} \right|_0^1$$

$$= \frac{1}{3} - \frac{1}{4}$$

$$= \frac{1}{12}$$



$\vec{F} = \nabla\phi$, then the line integral of $\vec{F}$ is independent of the path.

Q. If a force $\vec{F} = 2x^2y\hat{i} + 3xy^2\hat{j}$ displaces a particle in the xy -plane along the curve $y = 4x^2$ from $(0,0)$ to $(1,4)$, find the work done.

Solⁿ: Work done $= \int_C \vec{F} \cdot d\vec{r} = \int_C (2x^2y\hat{i} + 3xy^2\hat{j}) \cdot (dx\hat{i} + dy\hat{j})$

$$= \int_C 2x^2y dx + 3xy^2 dy$$

$C: y = 4x^2$
 $\Rightarrow dy = 8x dx, \quad 0 \leq x \leq 1.$

$\therefore$ work done $= \int_0^1 2x^2 \cdot 4x^2 dx + 3x(4x^2) \cdot 8x dx$

$$= \int_0^1 (8x^4 + 96x^4) dx = \int_0^1 104x^4 dx$$

$$= \left[\frac{104}{5} x^5 \right]_0^1 = \frac{104}{5}$$

Q. Show that $\int (y^2-1)dx + (z+zx+z^2)dy + (y+xy+2yz)dz$ is independent of the path of integration from $(1,2,2)$ to $(2,3,4)$. Evaluate the integral.

Solⁿ: Given integral is $\int \vec{F} \cdot d\vec{r}$ where $\vec{F} = (y^2-1)\hat{i} + (z+zx+z^2)\hat{j} + (y+xy+2yz)\hat{k}$.

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2-1 & z+zx+z^2 & y+xy+2yz \end{vmatrix} = 0$$

$\Rightarrow \vec{F}$ is irrotational $\Rightarrow \exists$ a scalar function ϕ such that

$$\vec{F} = \nabla\phi$$

$$\therefore \int \vec{F} \cdot d\vec{r} = \int \nabla\phi \cdot (dx\hat{i} + dy\hat{j} + dz\hat{k}) = \int \left(\frac{\partial\phi}{\partial x} dx + \frac{\partial\phi}{\partial y} dy + \frac{\partial\phi}{\partial z} dz \right)$$

$$= \int d\phi$$

$$= \phi \Big|_{(1,2,2)}^{(2,3,4)}$$

$$= \phi(2,3,4) - \phi(1,2,2)$$

Now to find ϕ .

$$\therefore F = \nabla \phi$$

$$(y_2 - 1)\hat{i} + (z + zx + z^2)\hat{j} + (y + xy + 2yz)\hat{k} = \frac{\partial \phi}{\partial x}\hat{i} + \frac{\partial \phi}{\partial y}\hat{j} + \frac{\partial \phi}{\partial z}\hat{k}$$

$$\frac{\partial \phi}{\partial x} = y_2 - 1 \quad ; \quad \frac{\partial \phi}{\partial y} = z + zx + x^2 \quad ; \quad \frac{\partial \phi}{\partial z} = y + xy + 2yz$$

① ② ③

$$\phi = xy_2 - x + A(y, z)$$

$$\frac{\partial \phi}{\partial y} = x_2 + \frac{\partial A}{\partial y}$$

comparing with ②, $\frac{\partial A}{\partial y} = \cancel{x_2} z + x^2$

$$\Rightarrow A = y_2 + y_2^2 + B(z)$$

$$\therefore \phi = xy_2 - x + y_2 + y_2^2 + B(z)$$

$$\frac{\partial \phi}{\partial z} = xy + y + 2yz + \frac{dB}{dz}$$

comparing with ③, $\frac{dB}{dz} = 0 \Rightarrow B$ is a pure constant
say $B = C_1$

$$\Rightarrow \phi = xy_2 - x + y_2 + y_2^2 + C_1$$

$$\int_C F \cdot d\vec{r} = \phi(2,3,4) - \phi(1,2,2)$$

$$= 82 - 15$$

$$= 67$$

Green's Lemma in the Plane

two functions $u(x,y)$ and $v(x,y)$ and their partial derivatives are single valued and continuous over a plane region R bounded by a closed curve C , then

$$\int_C u dx + v dy = \iint_R \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) dx dy.$$

Vector form:

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_R (\vec{\nabla} \times \vec{F}) \cdot \hat{k} dx dy$$

Q. Verify Green's Lemma in xy -plane for $\oint_C (xy+y^2) dx + x^2 dy$ where C is the closed curve of the region bounded by $y=x$ and $y=x^2$.

Solⁿ: Along $y=x^2$, $\int u dx + v dy$ is where $u=xy+y^2$, $v=x^2$ is

$$= \int_0^1 (x^3 + x^4) dx + x^2 \cdot 2x dx$$

$$= \int_0^1 (3x^3 + x^4) dx = \frac{3}{4} x^4 + \frac{x^5}{5} \Big|_0^1$$

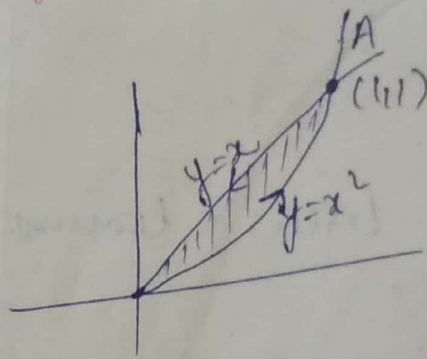
$$= \frac{3}{4} + \frac{1}{5} = \frac{19}{20}$$

Along $y=x$ from A to O

$$\int u dx + v dy = \int_1^0 (x^2 + x^2) dx + x^2 dx = \int_1^0 3x^2 dx = x^3 \Big|_1^0 = -1$$

$\therefore$ Given integral becomes $\frac{19}{20} - 1 = -\frac{1}{20}$

Now,



$$\iint_R \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) dx dy = \iint_R \left[\frac{\partial}{\partial x} (x^2) - \frac{\partial}{\partial y} (xy + y^2) \right] dx dy$$

$$= \iint_R (x - 2y) dx dy$$

$$= \int_{x=0}^1 \int_{y=x^2}^x (x - 2y) dy dx$$

$$= \int_0^1 \left. xy - y^2 \right|_{y=x^2}^x dx$$

$$= \int_0^1 (x^2 - x^2 - x^3 + x^4) dx$$

$$= \left. \frac{x^5}{5} - \frac{x^4}{4} \right|_0^1$$

$$= \frac{1}{5} - \frac{1}{4}$$

$$= -\frac{1}{20}$$

$\therefore$ Green's Lemma is verified.